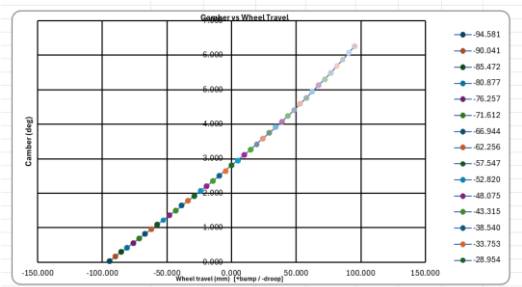


GOGO Kart Suspension Design

Front Suspension Config:

Suspension Arm Pivot Hardpoints (Inputs)			
Geometry Inputs	Value	Unit	Description
Y_U	289	mm	Upper inboard pickup axis lateral position (from origin)
Z_U	350	mm	Upper inboard pickup axis vertical position (from origin)
Y_L	274	mm	Lower inboard pickup axis lateral position (from origin)
Z_L	200	mm	Lower inboard pickup axis vertical position (from origin)
L_U	305	mm	Upper arm member length
L_L	328	mm	Lower arm member length
L_u	180	mm	Upright length (linear distance between upper and lower outboard rod end centers)
X_uf	360	mm	Longitudinal position of the upper front pickup (from origin)
X_ur	620	mm	Longitudinal position of the upper rear pickup (from origin)
X_lf	350	mm	Longitudinal position of the lower front pickup (from origin)
X_lr	610	mm	Longitudinal position of the lower rear pickup (from origin)
theta_U	0	deg	Upper arm angle from horizontal

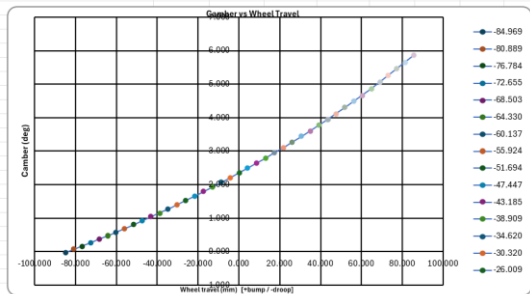
Calculated Final Values			
Parameter Name	Value	Unit	Description
caster	3.189	deg	side view angle of the steering axis from vertical
KPI	2.801	deg	(Kingpin inclination) front view angle between the steering axis and vertical
IC_det	-8141.305	mm^2	determinant of the two arm direction vectors (0 when the arms are parallel and increased when they diverge)
IC_y	-1249.461	mm	horizontal position of the instant center
IC_z	350	mm	vertical position of the instant center



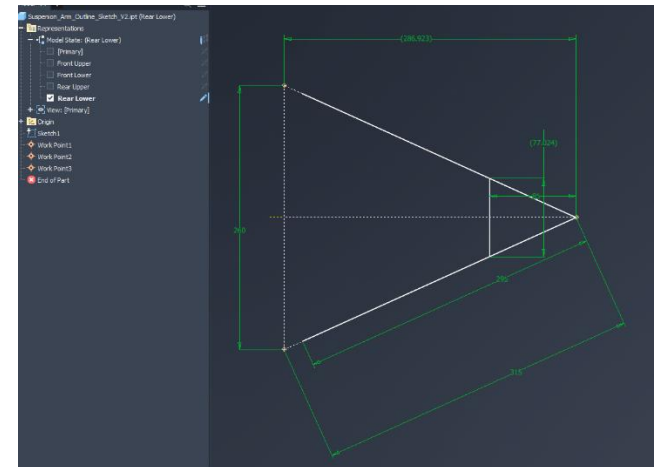
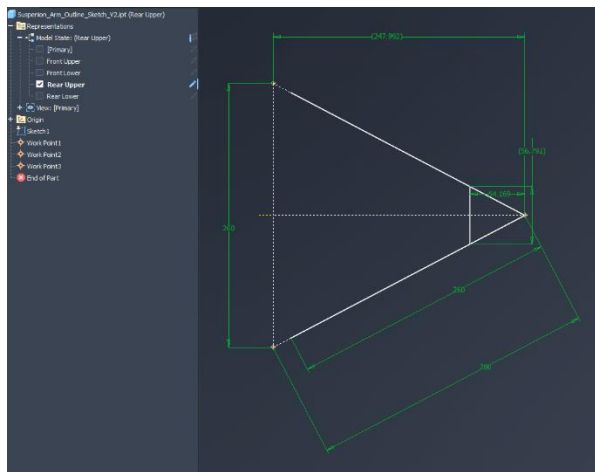
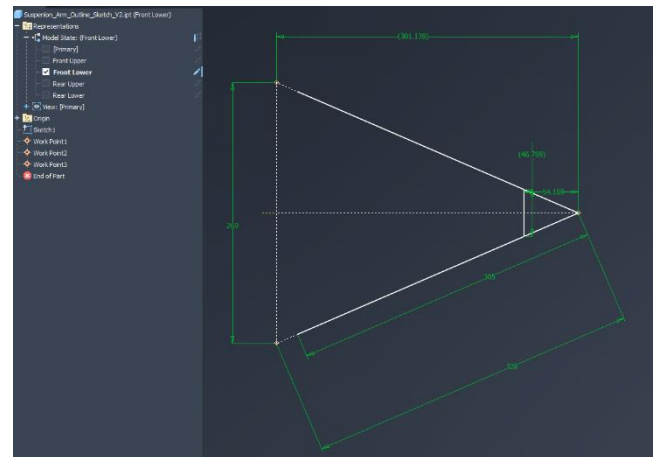
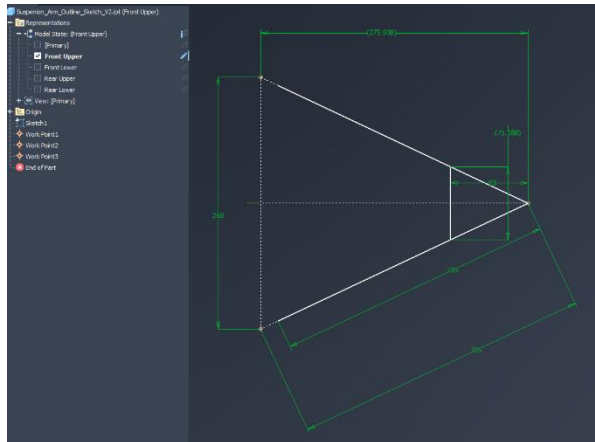
Rear Suspension Config:

Suspension Arm Pivot Hardpoints (Inputs)			
Geometry Inputs	Value	Unit	Description
Y_U	324	mm	Upper inboard pickup axis lateral position (from origin)
Z_U	350	mm	Upper inboard pickup axis vertical position (from origin)
Y_L	294	mm	Lower inboard pickup axis lateral position (from origin)
Z_L	200	mm	Lower inboard pickup axis vertical position (from origin)
L_U	280	mm	Upper arm member length
L_L	315	mm	Lower arm member length
L_u	180	mm	Upright length (linear distance between upper and lower outboard rod end centers)
X_uf	2180	mm	Longitudinal position of the upper front pickup (from origin)
X_ur	2440	mm	Longitudinal position of the upper rear pickup (from origin)
X_lf	2180	mm	Longitudinal position of the lower front pickup (from origin)
X_lr	2440	mm	Longitudinal position of the lower rear pickup (from origin)
theta_U	0	deg	Upper arm angle from horizontal

Calculated Final Values			
Parameter Name	Value	Unit	Description
caster	0.000	deg	side view angle of the steering axis from vertical
KPI	2.340	deg	(Kingpin inclination) front view angle between the steering axis and vertical
IC_det	-7402.279	mm^2	determinant of the two arm direction vectors (0 when the arms are parallel and increased when they diverge)
IC_y	-1140.057	mm	horizontal position of the instant center
IC_z	350	mm	vertical position of the instant center



A-Arm Sketches:



Mild Steel Rod Member Lengths (mm):

Front Upper: 285

Front Lower: 305

Rear Upper: 260

Rear Lower: 295

Suspension Kinematics Member Lengths (with rod end, mm):

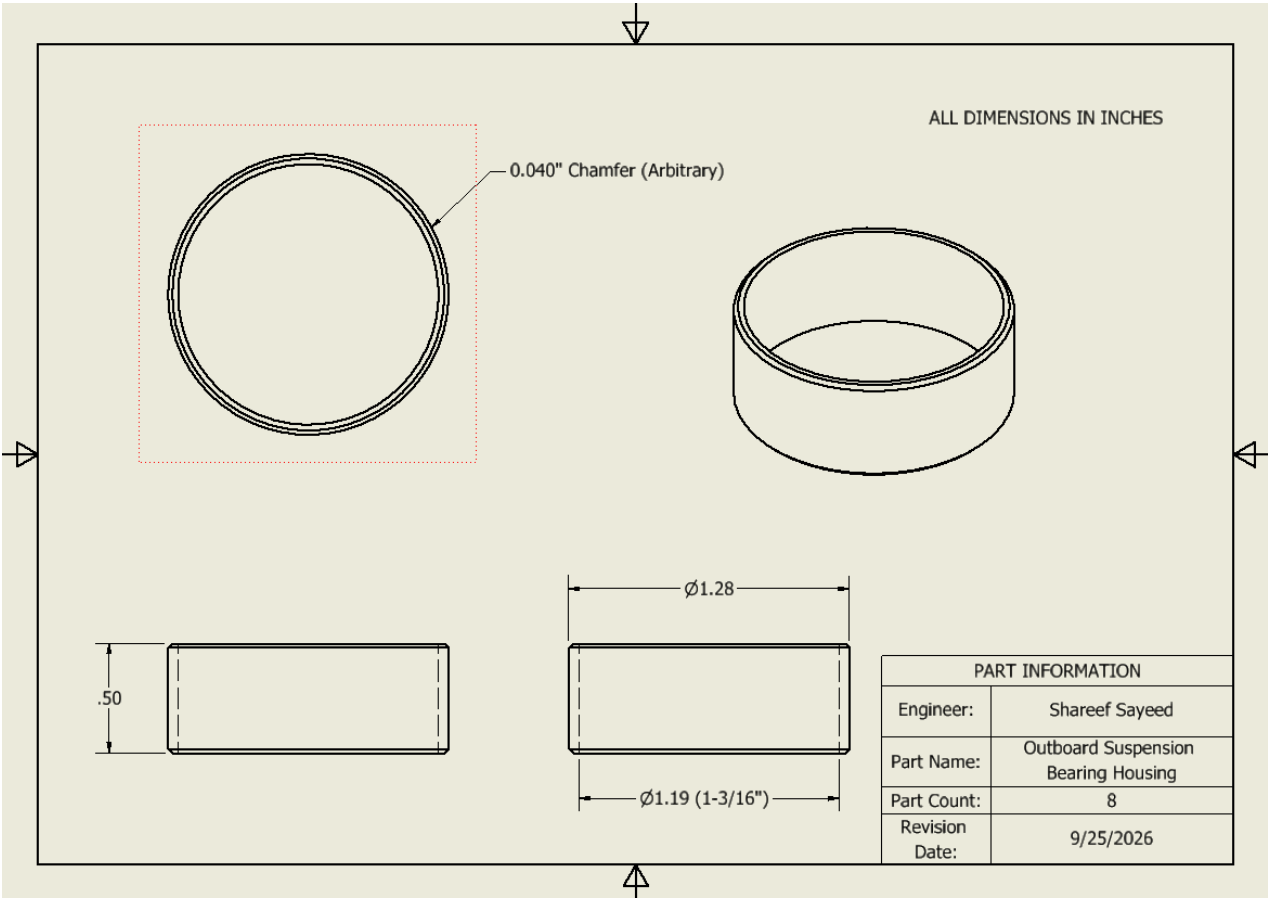
Front Upper: 305

Front Lower: 328

Rear Upper: 280

Rear Lower: 315

Outboard Bearing Housing:

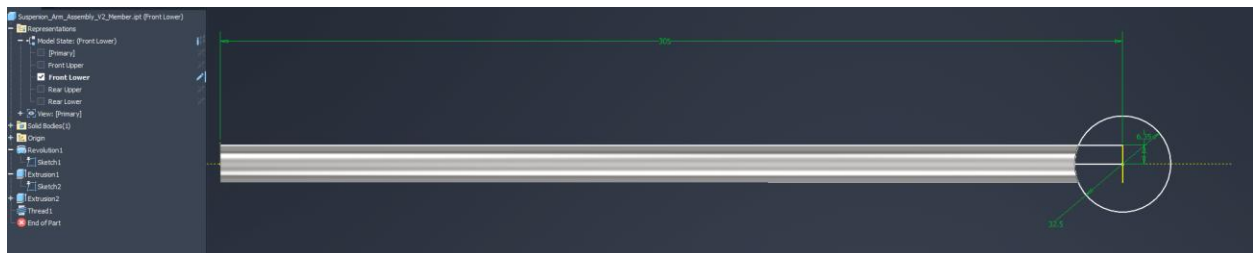


Suspension Members:

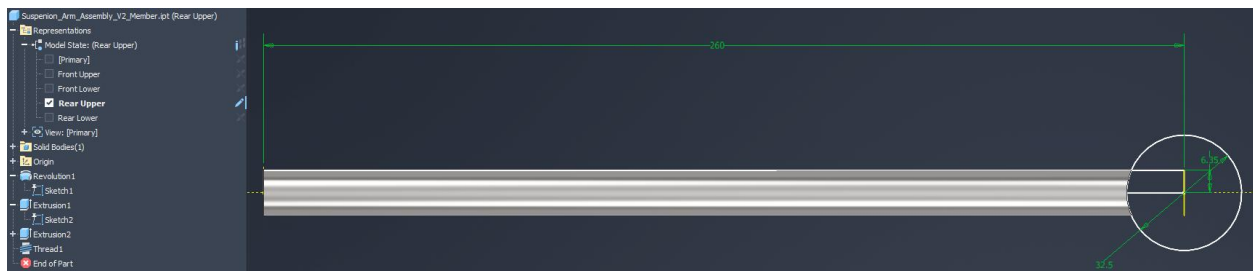
4x Front Upper Members (285mm/11.22" Length, 32.5mm/1.28" Cut Diameter)



4x Front Lower Members (305mm/12.01" Length, 32.5mm/1.28" Cut Diameter)



4x Rear Upper (260mm/10.02" Length, 32.5mm/1.28" Cut Diameter)



4x Rear Lower (295mm/11.61" Length, 32.5mm/1.28" Cut Diameter)

